

Chapter 157 — Diagnosis and Staging of Prostate Cancer — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **ERSPC** (European Randomised Study of Screening for Prostate Cancer): **20% ↓ cancer mortality**, 16 yr; 570 invited + 18 extra diagnoses per death prevented
- **PLCO** (Prostate, Lung, Colorectal, Ovarian): **no mortality difference** — contamination, low power
- **USPSTF** (US Preventive Services Task Force) 2018: **grade C, 55–69, shared decision**; none ≥70
- **Prostate disease** = main PSA (prostate-specific antigen) determinant; organ- not cancer-specific
- **5α-reductase inhibitor (5ARI) ≥12 mo** → **PSA × 2** (× 2.3 at 2 yr, × 2.5 at 7 yr)
- **Repeat raised PSA before biopsy**; no empirical antibiotics
- **PSA density (PSAD) ≥0.15; velocity (PSAV) >0.75 ng/mL/yr; % free PSA (%fPSA) ≤25%** — all at PSA 4–10
- **Gleason 3+4 = grade group (GG) 2; 4+3 = GG3** — primary pattern decides
- **D'Amico low** = T1–2a + PSA ≤10 + Gleason ≤6 (all three)
- **Low/intermediate risk** → **no routine CT or bone scan**; high → bone scan + CT/mpMRI

DIAGNOSIS

Screening

- **Since PSA era**: mortality ↓ >40%; advanced stage at diagnosis ↓ ~75%
- **Highest evidence**: randomised screened-vs-unscreened trials
- **ERSPC, PLCO**: both began 1993; endpoint cancer-specific mortality
- **Overtreatment worst >65**; mean age at diagnosis ~66
- **USPSTF**: 2008 insufficient ≤75, against ≥75 → **2012 grade D** → **2018 grade C**
- **ACS** (American Cancer Society): from **50** if life expectancy ≥10 yr; earlier if Black or first-degree relative diagnosed <65
- **AUA** (American Urological Association): shared decision **55–69**; 2-yearly interval
- **NCCN** (National Comprehensive Cancer Network): **baseline PSA at 45**; retest 1–4 yr
- **ACPM** (American College of Preventive Medicine): insufficient evidence for population screening

Digital rectal examination (DRE)

- **Limited value at PSA <3.0 ng/mL**; fair reproducibility only
- **DRE barely moves PSA** — within assay error
- **Standard practice**: PSA + DRE together

PSA

- **Kallikrein family**; free + complexed (α1-antichymotrypsin) = total (tPSA)
- **FDA** (Food and Drug Administration) approval for early detection: **1994**
- **+4% PSA per mL prostate volume**; volume ~30%, age ~5% of variance
- **Lower PSA**: hypogonadism, obesity, statins, 5ARI, BPH (benign prostatic hyperplasia) surgery
- **Finasteride 1 mg = 5 mg** for PSA lowering
- **5ARI**: baseline PSA first; or use nadir as new baseline
- **Ejaculation equivocal** → repeat after **48 h abstinence**
- **Black people**: higher PSA without cancer
- **Malmö**: PSA in 40s predicts metastasis/death >25 yr later
- **PCPT** (Prostate Cancer Prevention Trial): risk **continuous** — no safe PSA

Biopsy thresholds

- **4 ng/mL** original limit; **2.5–3** commonly used; **ERSPC 3**
- **>70 yr**: higher cut-off, e.g. **10 ng/mL**
- **Table 157.1**: PSA 0–1 → ~9% biopsy yield; >10 + suspicious DRE → ~50% high-grade

Derivatives and markers

- **%fPSA 25% ≈ PSAD 0.078** specificity — no volume needed
- **%fPSA unaffected** by race or 5ARI
- **cPSA** (complexed): specificity > tPSA, ≈ %fPSA
- **PSAV lethal risk**: >0.35 ng/mL/yr 10–15 yr pre-diagnosis (~5×); >2 ng/mL/yr in year pre-diagnosis
- **PSAV lower threshold**: 0.4 ng/mL/yr at low tPSA
- **bPSA** = BPH marker; **proPSA** ↑ in cancer; **hK2** (human kallikrein 2) tracks grade/volume
- **phi** (Prostate Health Index) = tPSA + fPSA + [–2]proPSA; FDA-approved
- **4Kscore** = tPSA + fPSA + intact PSA + human kallikrein 2
- **Urine**: SelectMDx (HOXC6, DLX1); ExoDx (3-gene exosomal); PCA3; TMPRSS2:ERG; MPS = tPSA + PCA3 + TMPRSS2:ERG
- **IsoPSA** (blood): validated ≥50 yr, PSA ≥4
- **Repeat-biopsy decision**: PCA3; **ConfirmMDx** (tissue methylation; negative predictive value >90%)
- **tPSA alone rarely triggers biopsy**

STAGING

Stage

- **Pathological stage** more accurate than clinical; survival inversely related
- **Post-RP (radical prostatectomy) predictors**: grade, margins, extracapsular extension, SVI (seminal vesicle invasion), nodes
- **TNM** (tumour, node, metastasis): American Joint Committee on Cancer 1975; Whitmore–Jewett historical
- **T1a ≤5%, T1b >5%** of resected tissue; **T1c** = needle biopsy, impalpable
- **T2a ≤½ one side; T2b >½ one side; T2c both**
- **T3a** extraprostatic; **T3b SVI**; **T4** fixed/other adjacent structures
- **M1a** non-regional nodes; **M1b** bone; **M1c** other ± bone
- **Prognostic groups**: TNM + PSA + grade group

Predicting extent

- **Organ-confined**: ~80% PSA <4; 66% 4–10; <50% >10
- **Nodes**: ~20% PSA >20; 75% >50
- **Poorly differentiated** → less PSA per gram
- **Prostatic acid phosphatase** obsolete
- **Abnormal DRE** → ↑ grade group 4–5

Grade

- **Gold standard grading**: Gleason (1966); grade = most important biopsy finding
- **Score** = primary + secondary pattern, 2–10; **2–5 no longer assigned**
- **Tertiary 5** on biopsy: 3+4 → 3+5; 4+3 → 4+5

- **GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10**
- **Adverse biopsy features:** intraductal, cribriform, SVI, periprostatic fat
- **Seminal vesicle biopsy:** only for large palpable basal tumour

Risk groups

- **D'Amico intermediate:** T2b or PSA >10–20 or Gleason 7
- **D'Amico high:** T2c or PSA >20 or Gleason 8–10
- **10-yr disease-free after RP:** 83% / 46% / 29%
- **Also validated:** CAPRA (Cancer of the Prostate Risk Assessment)

Imaging

- **mpMRI** (multiparametric MRI) = T2 + diffusion-weighted + dynamic contrast
- **Prior negative biopsy → strongly consider MRI;** suspicious lesion → targeted biopsy
- **PSMA PET** (prostate-specific membrane antigen): ⁶⁸Ga PSMA-11, ¹⁸F-DCFPyL — FDA-approved, initial staging, high risk
- **Staging imaging overused** in screened populations

Molecular staging and PLND

- **Prolaris** cell cycle; **Oncotype DX** 17 genes; **Decipher** 22 RNAs

- **PLND** (pelvic lymph node dissection): **no consistent oncological benefit** — staging only
- **Operate (PLND) if:** nomogram-predicted nodal risk exceeds guideline threshold

CLASSIC TRAPS

- **Trap:** PLCO proves screening useless — **Actually:** contaminated control arm, underpowered
- **Trap:** USPSTF says grade D — **Actually:** grade C since 2018 (55–69)
- **Trap:** PSA <4 excludes cancer — **Actually:** risk continuous; ~9% yield at 0–1
- **Trap:** Raw PSA on 5ARI — **Actually:** double it (≥12 mo)
- **Trap:** Low PSA in hypogonadism reassures — **Actually:** may mask cancer
- **Trap:** Antibiotics for raised PSA — **Actually:** not without urinary symptoms
- **Trap:** Gleason 7 is one group — **Actually:** 3+4 = GG2, 4+3 = GG3
- **Trap:** SVI is T4 — **Actually:** T3b
- **Trap:** Non-regional nodes are N1 — **Actually:** M1a
- **Trap:** Bone scan for all new cancers — **Actually:** high risk only